AKTU

B. Tech II-Year

Gateway Classes

Semester -IV

CS IT & Allied Branches

BCC402-PYTHON PROGRAMMING

UNIT-2 : Python Program Flow Control

Hand Written Notes





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BCC402- PYTHON PROGRAMMING

UNIT-2: Python Program Flow Control

Hand Written Notes

Syallbus

Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in python, For loop using ranges, string, list and dictionaries. Use of while loops in python, Loop manipulation using pass, continue, break and else. Programming using Python conditional and loop blocks.

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MODULE - 2

Conditionals and loops

Conditional statements

A conditional statement as the name suggest itself, is used to handle conditions in your program. These statements guide the program while making decisions based on the condition encountered by the program, we will explore three conditional statements. in python:

if statement, if -else statements, if -elif- else ladder.

Indentation in python

- For the ease of programming and to achieve simplicity, python doesn't allow the use of parentheses for the block level code. In python, indentation is used to declare a block. If two statements are at the same indentation level, then they are the part of the same block.
- Generally, four spaces are given to indent the statements which are a typical amount of indentation in python.

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Indentation is the most used part of the python language since it declares the block of code. All the statements of one block are intended at the same level indentation. we will see how the actual indentation takes place in decision making and other stuff in python.

The If Statement

The if statement is used to test a particular condition and if the condition is true, it executes a block of code known as if- block. The condition of if- statement can be any valid logical expression which can be either evaluated to true or false.

Syntax:

```
if expression:
    statement
```

Example:

```
num = int (input (" enter the number:"))
    if num%.2 == 0:
        print (" The given number is an even no")
```

Output:

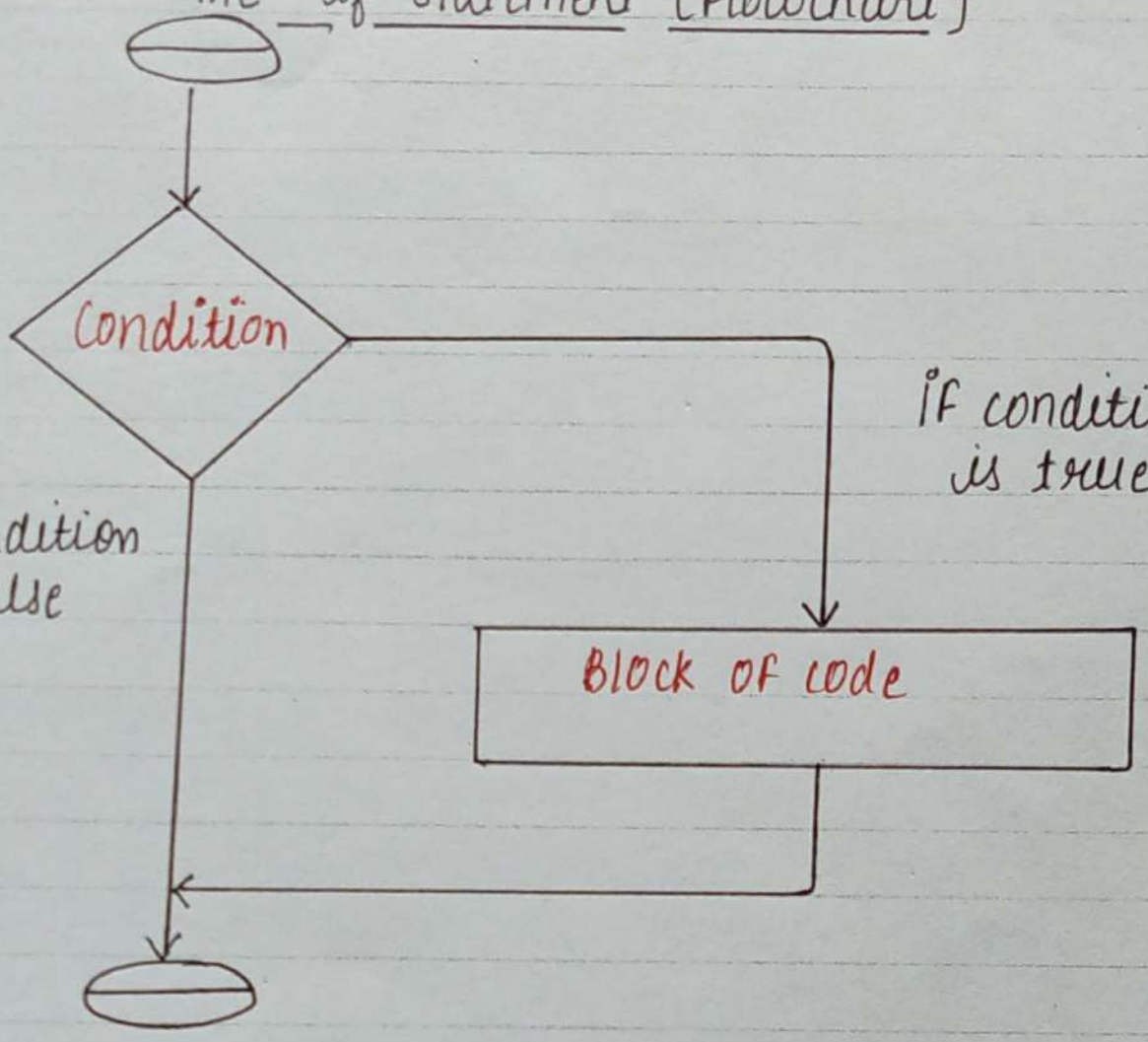
```
Enter the number 8
The given number is an even no.
```


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The if statement (Flowchart)



Example:

```
num = int( input ("enter the number:"))
if num<12:
    print ("hello")
if num >3 and num<12:
    print ("hi")
    print(" how are you?")
if num >4 and num <14:
    print ("bye")
if num>12:
    print ("jjj")
```

Output:

```
enter the number:5
hello
hi
how are you?
bye
```

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Write a program to find out the largest number among three

```
a= int [input( "Enter a:"))
b= int ( input ("Enter b:"))
c= int (input ("Enter c:"))
if a>b and a>c:
    print(" From the above three numbers
        given a is largest")
if b>a and b>c:
    print(" From the above three
        numbers given b is largest")
if c>a and c>b:
    print ("From the above three numbers
        given c is largest")
```

output

Enter a: 6
Enter b: 7
Enter c: 5

7 is the largest number


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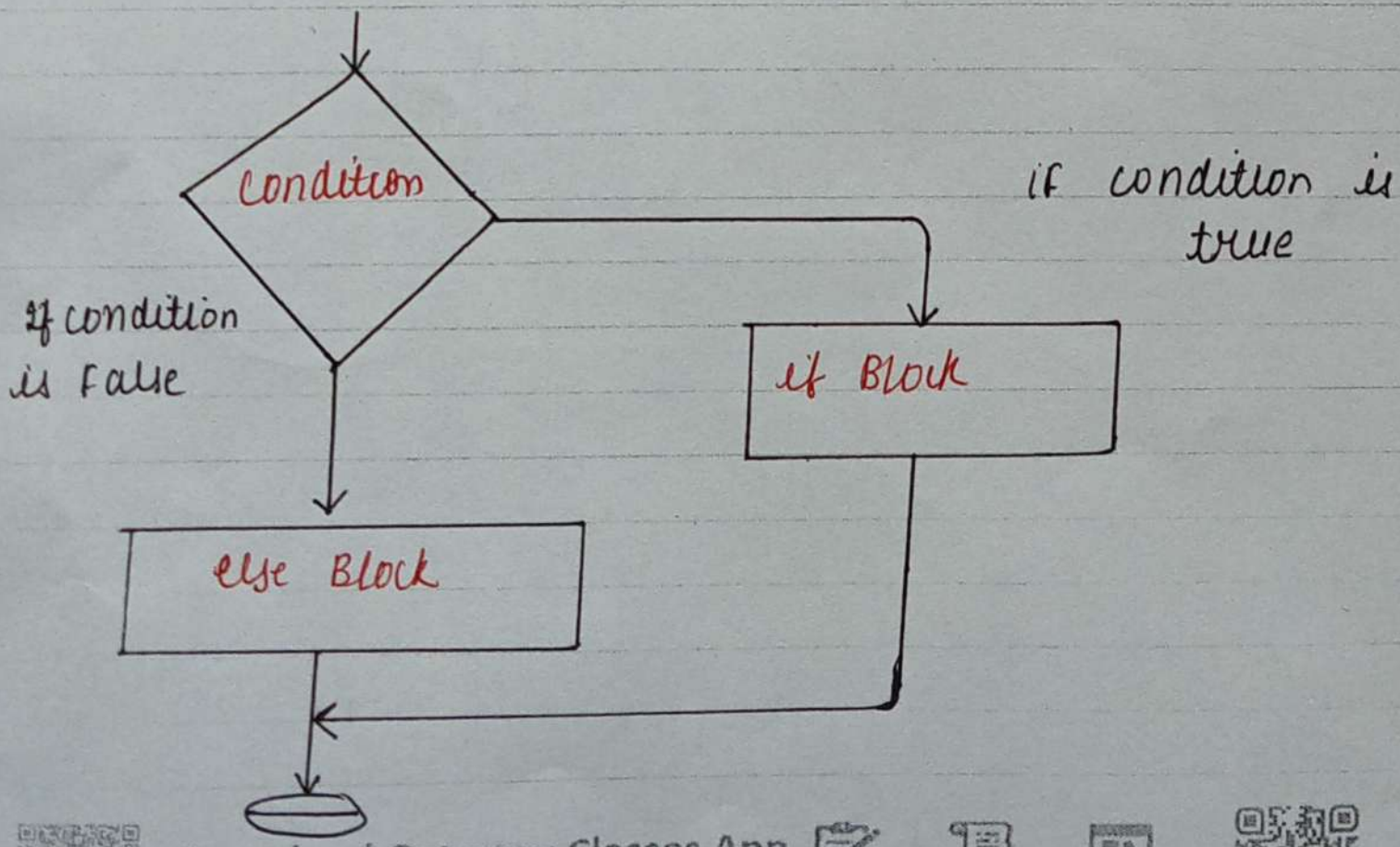
If else statement

The if-else statement provides an else block combined with the if statement which is executed in the false case of the condition. If the condition is true, then the if-block is executed. otherwise, the else-block is executed.

Syntax:

```
if condition:
    # block of statements
else:
    # another block of statements (else-block).
```

If-else statement Flowchart



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- Write a program to check whether you are eligible to vote or not.

```
age= int (input ( "Enter your age:"))
    if age>=18:
       print (" You are eligible to vote !!")
    else:
     print ( "You are not eligible to vote!!")
```

Output:

```
Enter your age :27
You are eligible to vote!!
```

- Write a program to check whether number is even or not.

```
num= int ( input(" enter your number:"))
    if  num%2==0:
       print (" even number")
    else:
     print ( "odd number")
```

Output:

```
6
even number
```


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- Write a program to check whether number is divisible by 3 or not.

```
num = int (input ("enter the no."))
if num %3==0:
    print ("The given number is divisible by 3")
    else:
    print ("The given number is not divisible
                by 3")
```

output:

```
enter the number: 6
The given number is divisible by 3
```

The elif statement

The elif statement enables us to check multiple conditions and execute the specific block of statements depending upon the true condition among them. We can have any number of elif statements in our program depending upon our need. However, using elif is optional.

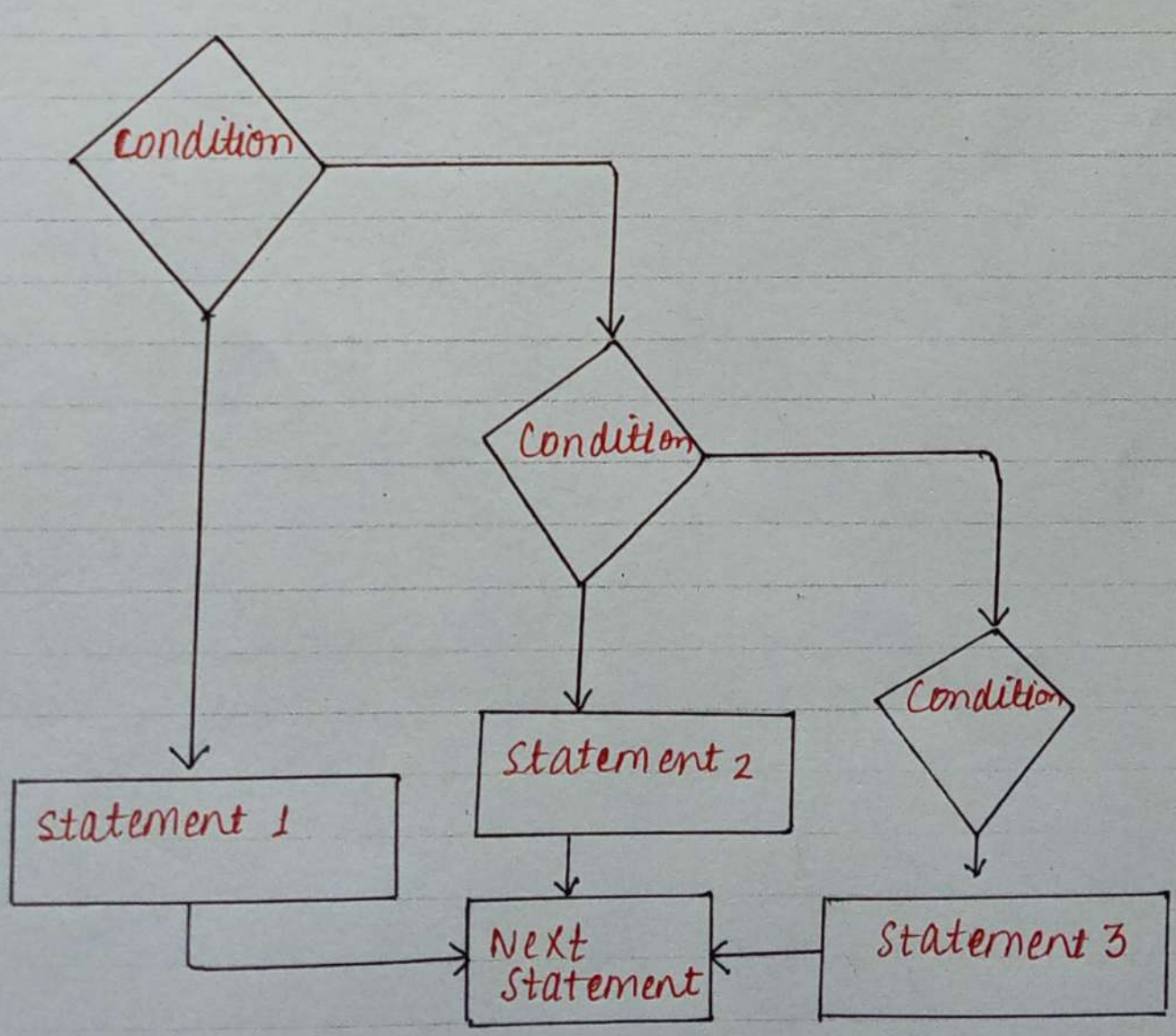
The elif statement works like an if-else-if ladder statement in C. It must be succeeded by an if statement.

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The syntax of the elif statement is given below

```
if expression 1:
    # block of statements
elif expression 2:
    # block of statements
elif statement 3:
    # block of statements
else:
    # block of statements
```

The elif statement (Flowchart)



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Example 1

```
number= int ( input ("Enter the number?"))
    if number == 10:
       print (" The given number is equals to 10")
    elif number== 50:
       print (" The given number is equal to 50")
     elif number==100:
       print (" The given number is equal to
                100")
    else:
    print (" The given number is not equal to
                10, 50 Or 100")
```

Output

```
Enter the number ?6
The given number is not equal to 10, 50
                or 100.
```

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Example 2:

```
marks = int ( input ("Enter the marks?"))
 if marks > 85 and marks<= 100:
   print (" congrats ! you scored grade A ...")
   elif marks > 60 and marks <=85 :
     print (" you scored grade B+ ..")
   elif marks >40 and marks<= 60 :
     print (" you scored grade B ..")
   elif ( marks> 30 and marks< =40):
     print (" You scored grade C ..")

    else :
     print ( " sorry you are Fail")s
```

output:

```
Enter your marks ? 70
 you scored grade B+ ---
```

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- Write a program to convert month name to a number of days

```
print(" List of months: January, February,
    March, April, May, June, July, August,
     September, October, November, December").
month_name = input(" Enter the name of
                    the month:")
    if month_name == " February ":
        print (" No. of days : 28/29 days")
elif month_name in(" April ", "June", " September",
                   "November"):
    print(" No. of days : 30 days")
elif month_name in(" January", "March", "May",
        "July" , "August ", " October ", " December"):
    print (" No. of days : 31 day")
else:
    print(" Wrong month name")
```

Output

List of months: January, February, March, April, May, June, July, August, September, October, November, December.

Enter the name of Month: April

No. of days : 30 days

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- Write a program to input a character and check whether a character is vowel and consonants.

```
ch= input(" enter the character\n")
if ch in ( "a", "e", "i", "o", "u", "A", "E",
           "I", "O", "U"):
    print (" vowels")
else:
    print(" consonants")
```

- Write a program to input a character and check whether a character is vowel and consonants or digit and special symbol.

```
ch= input(" enter the character\n")
if ch in ("a", "e", "i", "o", "u", "A", "E"
          "I", "O", "U"):
    print (" vowels")
elif ch >= " A" and ch <= "z" or ch >= "a"
        and ch <= "Z":
    print (" consonants")
elif ch >= "0" and ch <= " 9":
    print (" digit")
else:
    print (" special symbols")
```


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- Write a program to check whether is alphabet (uppercase and lower case and non - alphabet symbol)

```
ch = input(" enter the character \n")
if ch>"A" and ch<="Z":
    print(" upper case alphabet")
elif ch>="a" and ch<="z":
    print(" lower case alphabet")
else:
    print ("non - allphabet symbol")
```

- Write a program to enter the angle and check what type of triangle is

```
a= int( input(" enter angle"))
b = int (input(" enter angle"))
c= int (input("enter angle"))
if a+b+c==180 and a>0 and b>0 and c>0:
    print ("valid")
    if a==b and b==c and c==a:
        print (" equilateral")
    if a==b or b==c or c==a:
        print (" iso")
    if a==90 or b==90 or c==90:
        print ("right")
    else:
    print("invalid")
```

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- Write a program to enter the side and check which type of triangle is

```
a = int( input (" enter the first sides of triangle \n"))
b = int (input (" enter the second side of triangle \n"))
c = int( input (" enter the third side of triangle \n"))
    if a+b>c and b+c>a and c+a>b:
       print (" valid triangle")
     if a==b and b==c and c==a:
       print (" equilateral triangle"):
     if a==b or b==c or c==a:
       print (" isosceles triangle")
     if a!=b and b!=c and c!=a:
       print (" scalene triangle")
if a*a+b*b==c*c or b*b+c*c==a*a
          or a*a+c*c==b*b:
    print (" right angled triangle")

else:
    print (" invalid triangle")
```

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• Write a program to find out the largest number among three

```
a= int (input ("enter the first value\n"))
b= int (input ("enter the second value \n"))
c= int (input ("enter the third value\n"))
  if a>b and a>c:
        print (a," is the largest")
  elif b>a and b>c:
     print (b," is the largest")
     else:
       print (c, " is the largest")
```

output :

```
enter the first value
 4

enter the second value
  5

  enter the third value
   6

6 is the largest
```

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- Write a program to check which is largest among three using nested if-else

```
a = int ( input ( "enter the value of a\n"))
  b = int ( input ("enter the value of b\n"))
    c = int ( input ("enter the value of c\n"))
      if a>b:
          ~~pri~~
           if a>c:
             print (a," is the largest")
        else:
          print (c," is the largest")

      else:
          if b>c:
             print (b," is the largest")
          else:
             print ( c, is the largest")
```

'Write a program to check whether year is leap or not

```
x= int(input ("enter the year\n"))
if x %.400==0:
   print (x," is leap year")
elif  x %. 100==0:
   print (x," is not a leap year")
elif  x%.4 ==0:
   print (x, "is leap year")
  else:
   print (x, " is not leap year")
```


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- Write a program to find out the roots of quadratic equation

```
import math
print(" enter the coefficient of quadratic equation")
a = float( input(" enter the x^2 coefficient \n"))
b = float ( input ("enter the X coefficient \n"))
c = float (input ("enter the X^0 coefficient "))
   d = b * b - 4 * a * c
    sqrt_val = math. sqrt (abs(d))
      if d>0:
        print ("Roots are real and distinct")
     r1 = (-b+ sqrt_val ) / (2*a)
     r1 = (-b- sqrt_val) / (2 * a)
      print ( " Root 1 =", r1 )
      print ( "Root 2 =" , r2)
elif d==0:
      print ("roots are equal")
       r1 = r2 = -b / (2*a)
      print ("root 1 =", r1 )
       print ("root 2 =", r2 )
   else:
      print ("roots are imagenary")
        real = - b/ (2*a)
      imaginary = sqrt_val / (2 * a)
      print( real, "+i", imaginary)
      print (real , "-i", imaginary)
```


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- Write a program to make calculator using elif statement.

```
x = float (input ("enter the first number\n"))
y = float ( input ("enter the second number\n"))
z = input(" enter the operator (+, - , * , / , //) \n")
  if  z == " + ":
        w = x + y
       print ( w, " the sum of two number")
     elif  z == " - ":
          w = x - y
         print ( w , " substraction of x - y number")
    elif  z == " * ":
        w = x * y
        print (  w, " multiplication of two
                    number")
    elif  z == " / ":
        w = x / y
        print (  w, " division of x / y ")
elif  z == "//":
    w = x // y
    print  (w, " Floor division of x // y")
    else:
    print (" enter the  wrong choice")
```


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Match Case

For developers coming from languages like C / C++ or Java know that there was a conditional statement known as Switch Case. This Match-Case is the switch case of Python which was introduced in python 3.10 Here we have to first pass a parameter then try to check with which case the parameter is getting satisfied. If we find a match we will do something and if there is no match at all we will do something else.

```
parameter = " Geekforgeeks"
    match parameter:

        case first :
            do_ something (First)
    case second:
        do_ something ( second);

    case third:
            do_ something (third)
            ____
            ____

    case n:
            do_ something (n)
    case _:
            nothing_ matched _function ()
```

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Write a program to print the day in a week using match case

```
print (" enter 1- monday, 2 - tuesday, 3- wednesday,
        4 - thursday" , 5 - Friday , 6 - saturdg
            7- sunday")
    x = input ("enter the number")
      match x:
          case "1":
              print (" monday")
          case "2":
              print (" tuesday")
           case "3":
                print (" wednes day")
           case "4":
                (print (" thursday")
          case "5":
              print (" friday")
          case "6":
              print (" saturday")
           case "7":
              print (" sunday")
      case_:
        print ( " wrong choice enter again!")
```

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* Write a program to check whether is a vowel or consonants.

```
X = input ("enter the string of length 1\n")
    match x:
        case "A":
            print ("vowel")
        case "E":
            print ("vowel")
        case "I":
            print (" vowel")
        case "O":
            print (" vowel",
        case "U":
            print (" vowel")
        case"a":
            print (" vowel")
        case" e":
            print (" vowel")
        case" i":
            print ("vowel")
        case "o":
            print (" vowel")
        case"u":
            print (" vowel")
        case"u":
            print ("vowel")
        case _ :
            print(" consonants")
```

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- Write a program to print the number of month in a year

```
print (" 1 - Jan, 2 - Feb, 3 - Mar, 4 - Apr, - 5 - May, 6 - June,
        7 - July, 8 - Aug, 9 - Sep, 10 - Oct, 11 - Nov,
        12 - Dec ")
x = input (" enter the number for displaying the
        particular year")
match X:
    case "1":
        print (" Jan")
    case "2":
        print (" Feb")
    case "3":
        print ( "Mar")
    case "4":
        |
        |
        |
        |
        |
        |
    case "11":
        print ("Nov")
    case "12":
        print ("Dec")
    case _:
        print (" Invalid choice enter again")
```


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Write a program to make calculator using match case:

```
x = Float( input ("enter the first number\n"))
y = Float ( input (" enter the second number \n"))
z = input (" enter the operator(+ , * , / , - // ) \n")
 match z :
     case " + ":
       w = x + y
        print ( w, " the sum of two number")
     case " - ":
         w = x - y
         print( w, " subtraction of x - y number")
     case " * ":
         w = x * y
         print  (w, " multiplication of two number")
 case  " / ":
         w = x / y
         print (w, " division of x/y")
 Case " // ":
         w = x // y
         print ( w, " floor division of x//y")
     case _ :
         '
         print (" enter the wrong choice")
```


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Loops

➡ A loop is a programming structure that repeats a sequence of instructions until a specific condition is satisfied.

➡ A loop statement allows us to execute a statement or group of statements multiple times Python programming language provides following types of loops to handle looping requirement.

- while (entry controlled loop)
- for
- nested

while loop

In python, a while loop is used to execute a block of statements repeatedly until a given condition is satisfied. And when the condition becomes false, the line immediately after the loop in the program is executed.

Syntax:

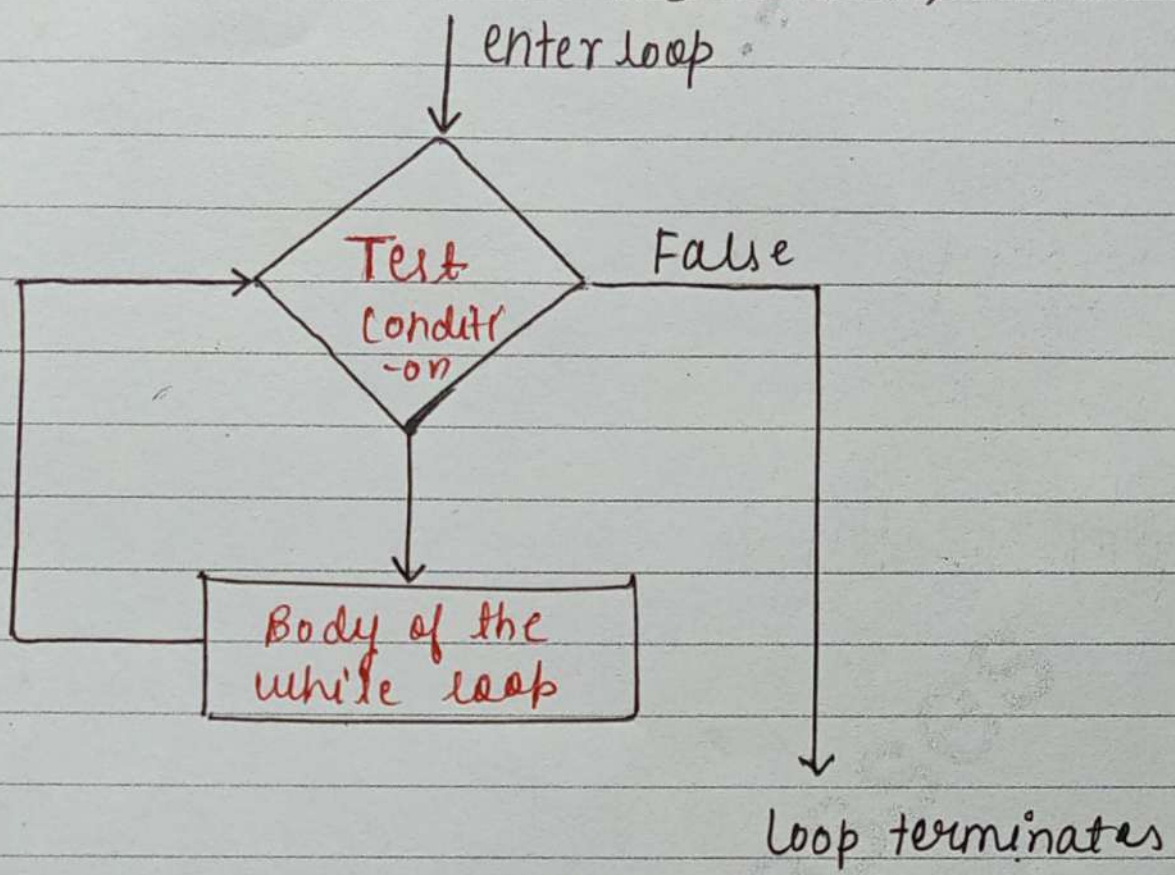
```
while expression:
    statement
```

All the statements indented by the same number of character spaces after a programming construct are considered to be a part of single block of code.

Python uses indentation as its method of grouping statements.


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WHILE LOOP (FLOWCHART)



while loop

```
x = int( input (" enter the number \n"))
while x < 11 :
        print ("hello")
        x = x - 1
```

Output

```
enter the number
    6
    hello
    hello
    hello
    hello
    hello
```


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Write a program to print first n natural number

```
x = int(input("enter the number \n"))
i = 1
while (i<=x):
    print(i)
    i=i+1
```

Write a program to print the sum of first n natural number

```
x =int( input(" enter the number \n"))
num=0
i=1
while (i<=x):
    print(i)
    sum= sum+i
    i=i+1
print(" sum=", sum)
```

Write a program to reverse a n digit number

```
n =int(input(" enter n digit number \n"))
rev=0
while n>0:
    d=n%10
    rev= rev*10+d
    n=n//10
print(" reverse", rev)
```


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Write a program to count the number of digit in a n digit number

```
num = int( input(" enter the number "))
    count = 0
        while num > 0:
            num = num//10
            count = count +1
    print (" Number of digits :", count)
```

Write a program to check whether a number is pallindrome or not

```
n = int( input  (" enter n digit number \n"))
    num = n
     rev = 0
     while n > 0:
          d = n % 10
          rev = rev * 10 + d
          n = n//10
     print (" reverse ", rev)
     if rev == num:
          print ( " pallindrome")
     else:
      print (" Not pallindrome")
```

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write a program to check whether a number is Armstrong or not

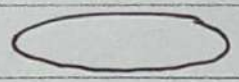
```
n = int (input (" enter n digit number\n"))
num = n
sum = 0
while n > 0:
    d = n % 10
    sum = sum + ( d * d * d )
    n = n // 10
if sum == num:
    print ("Armstrong")
else:
    print ("not armstrong")
```

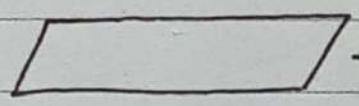

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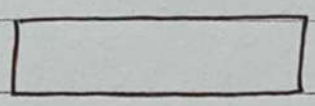
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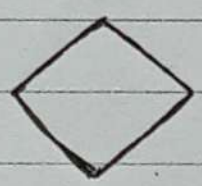
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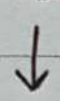
Flowcharts & Algorithms...

 — oval — start / end

 — Parallelogram Print / input

 — Rectangle Processing

 — Diamond Condition

 — arrow connect two or more symbol

Flowchart symbol

Write a program to print first n natural number

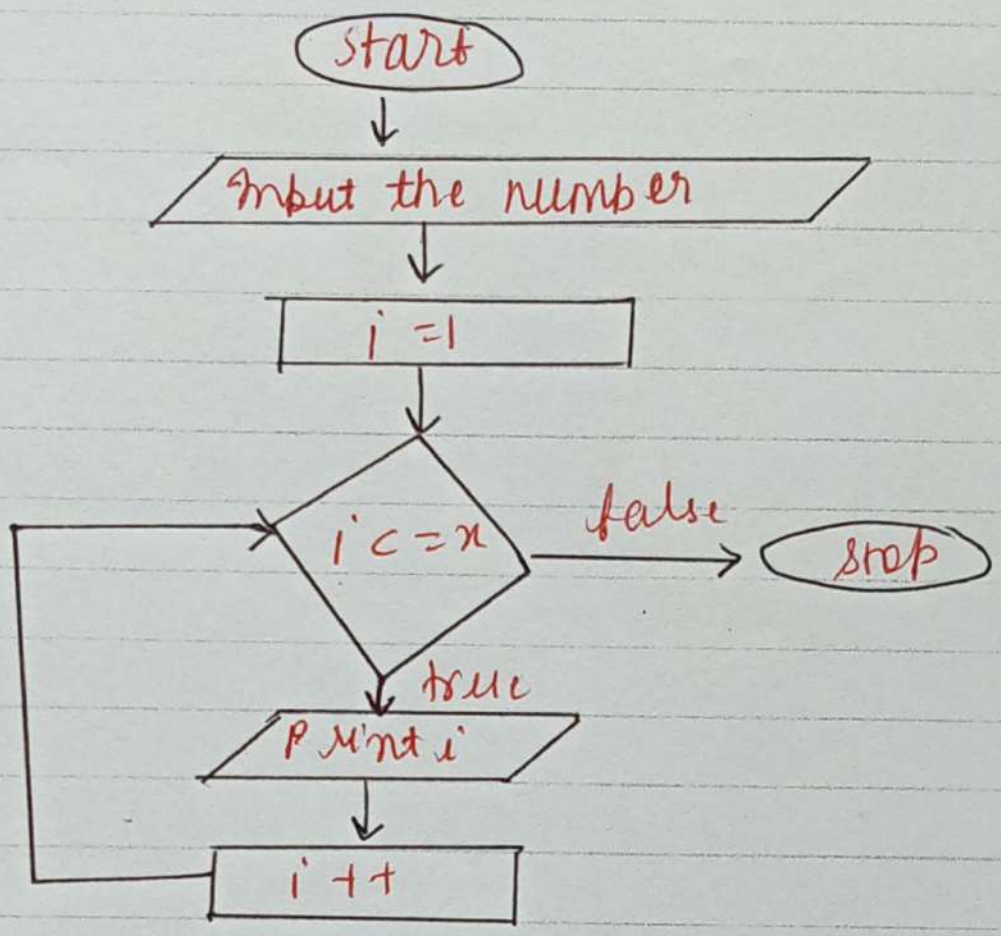
```
x = int(input("enter the no.\n"))
i=1
while (i<=x):
    print (i)
    i=i+1
```

Algorithm

step.1 start
step.2 enter the number
step.3 i=1
step.4 while <=x. then go to step 5 otherwise go to step 7.
step.5 Print i
step.6 i=i+1 then go to step 4
step.7 Stop

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Flowchart



Write a program, algorithm and flowchart to print the sum of first of n natural numbers.

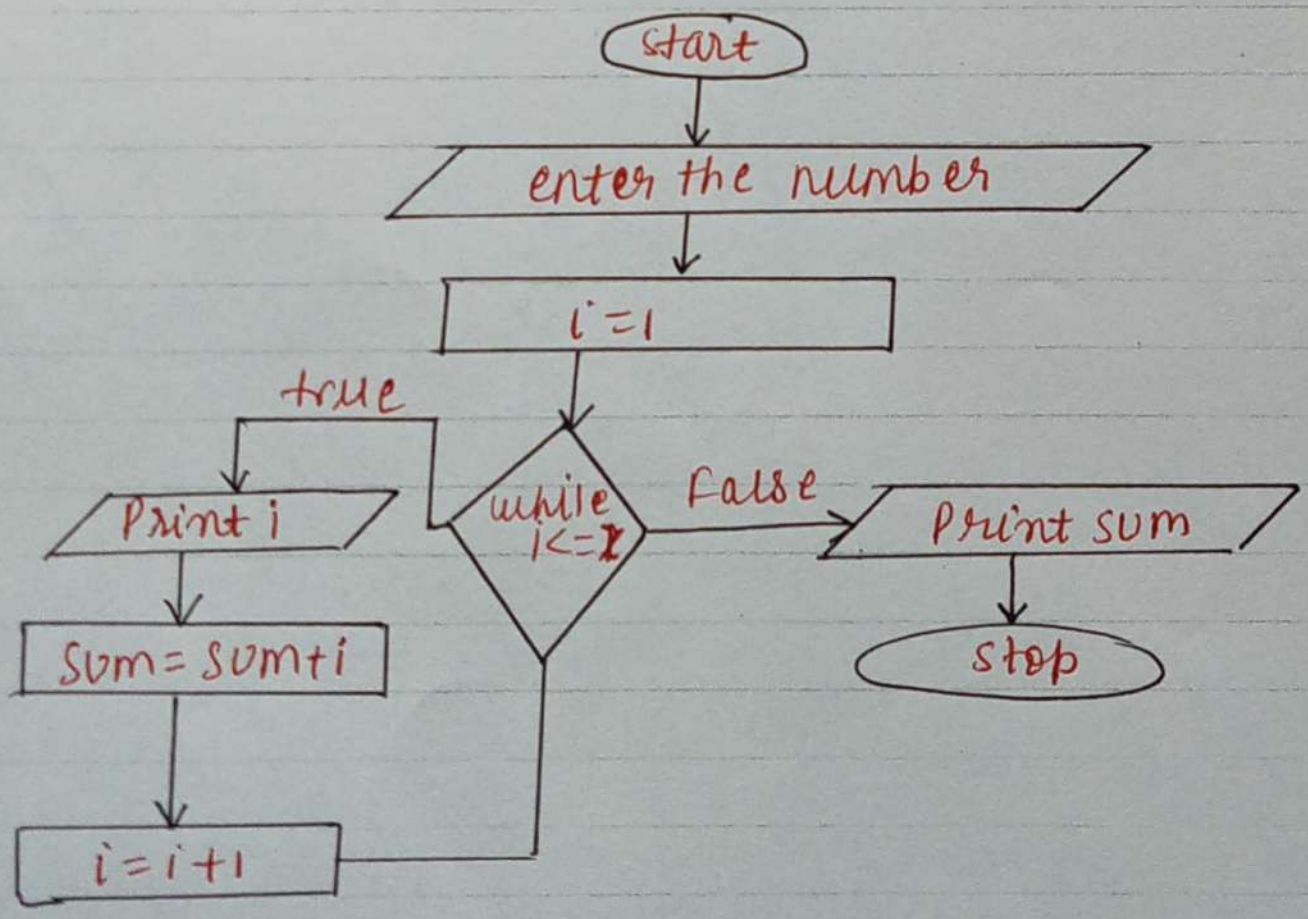
```
x = int (input(" enter the number \n"))
    sum = 0
     i=1
    while ( i<=x):
          print (i)
          sum = sum+i
           i=i+1
        print (" sum =", sum)
```

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Algorithm

step 1. Start
step 2 enter the number
step 3 i=1
step 4 while i<=x true go to step 5 otherwise go to step 8
step 5 print i
step 6 sum = sum+i
step 7 i=i+1 , then go to step-4
step 8 print sum
step 9 stop

Flowchart



Note: Flowchart is the pictorial representation of Algorithm.

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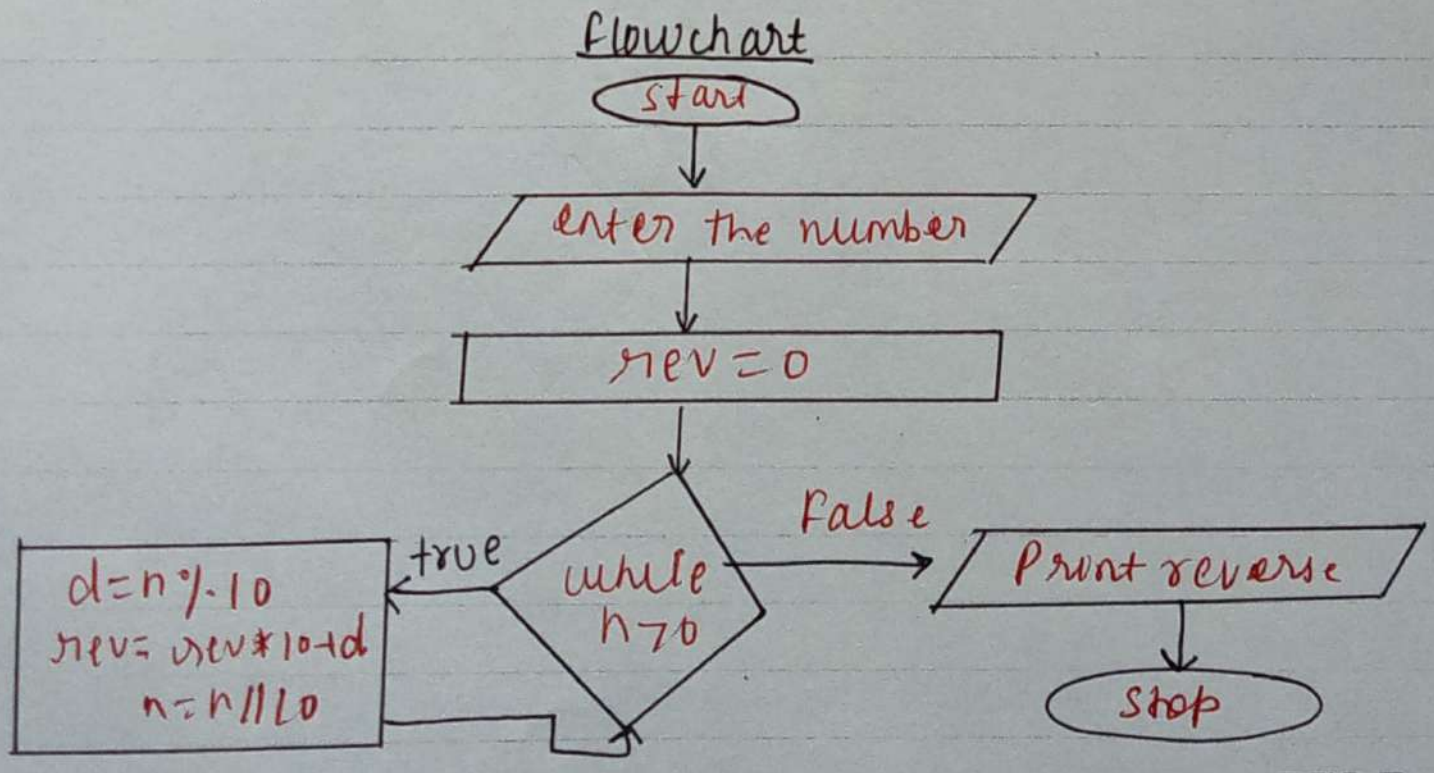
Write a program to reverse a n digit number

```
n=int(input("enter n digit number\n"))
rev=0
while n>0:
    d=n%10
    rev=rev*10+d
    n=n//10
print("reverse",rev)
```

Algorithm

step 1. start
2. enter a n digit number
3. rev=0
4. while n>0 true, go to step 5 otherwise go to step 6.
5. d=n%10, rev=rev*10+d, n=n//10, go to step 4
6. print reverse
7. stop

Flowchart



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Write a program to check whether a number is pallindrome or not

```
n= int(input(" enter n digit number\n"))
    num=n
    rev=0
    while n>0:
      d= n%.10
      rev=rev*10+d
        n=n//10
      print(" reverse", rev)
    if rev==num:
        print (" pallindrome")
  else:
    print(" Not pallindrome")
```

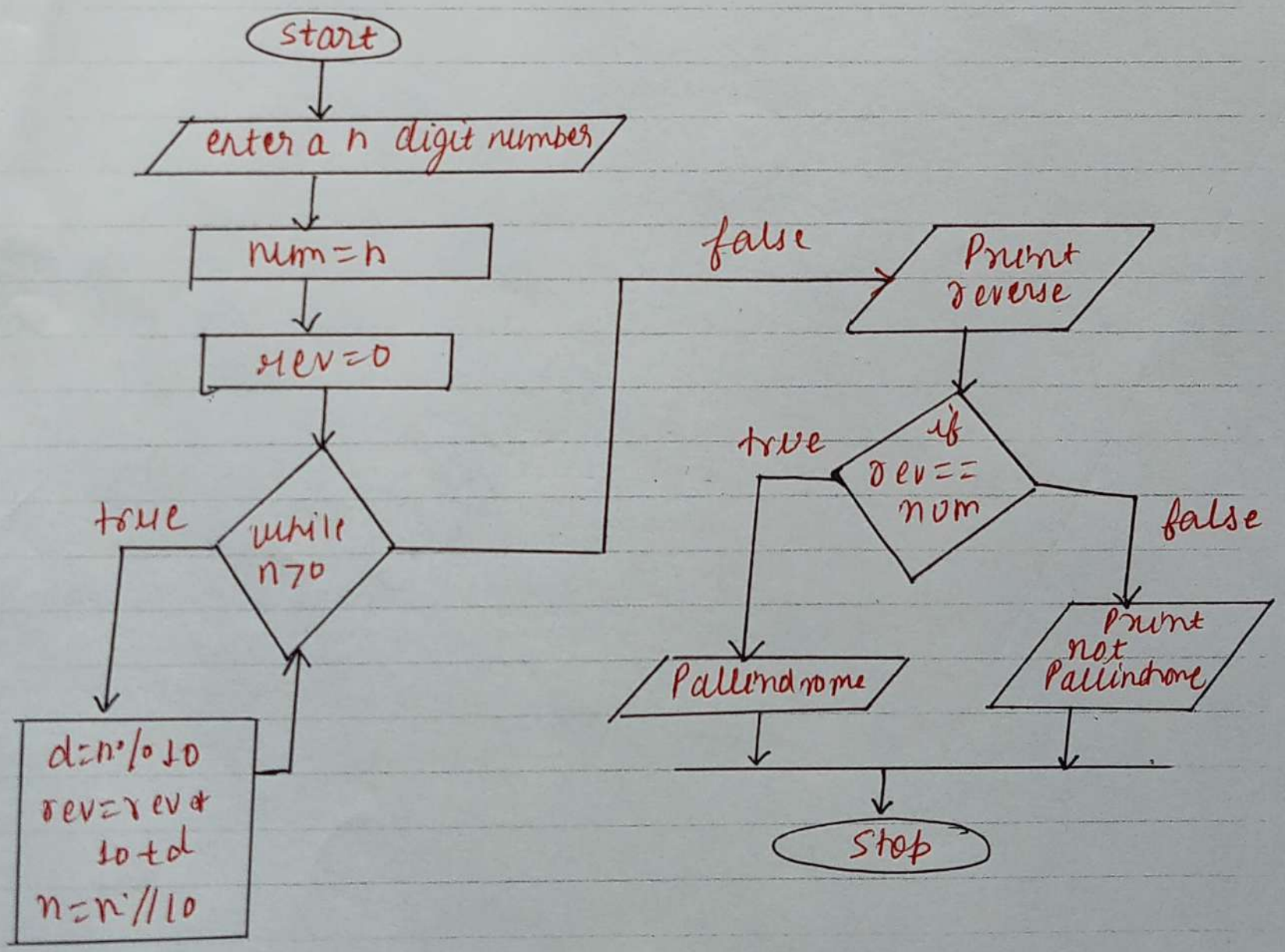
Algorithm

1. Start
2. enter a n digit number
3. num=n
4. rev=0
5. while n>0 true,
   then go to step 6 otherwise go to step 7.
6. d= n%.10
   rev= rev*10+d
   n=n\\10
   then go to step 5

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7. print rev
8. if (rev == num) . true, go to step 9 else step 10
9. Print pallindrome , then go to step 11
10. print not pallindrome
11. Step.

Flowchart



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Write a program to check whether a number is Armstrong or not

```
n = int( input ( " enter n digit number\n"))
  num=n
   sum=0
   while n>0:
        d=n%.10
          sum = sum+( d*d*d )
           n=n//10
     if sum == num:
          print (" Armstrong ")
     else:
       print (" not Armstrong ")
```

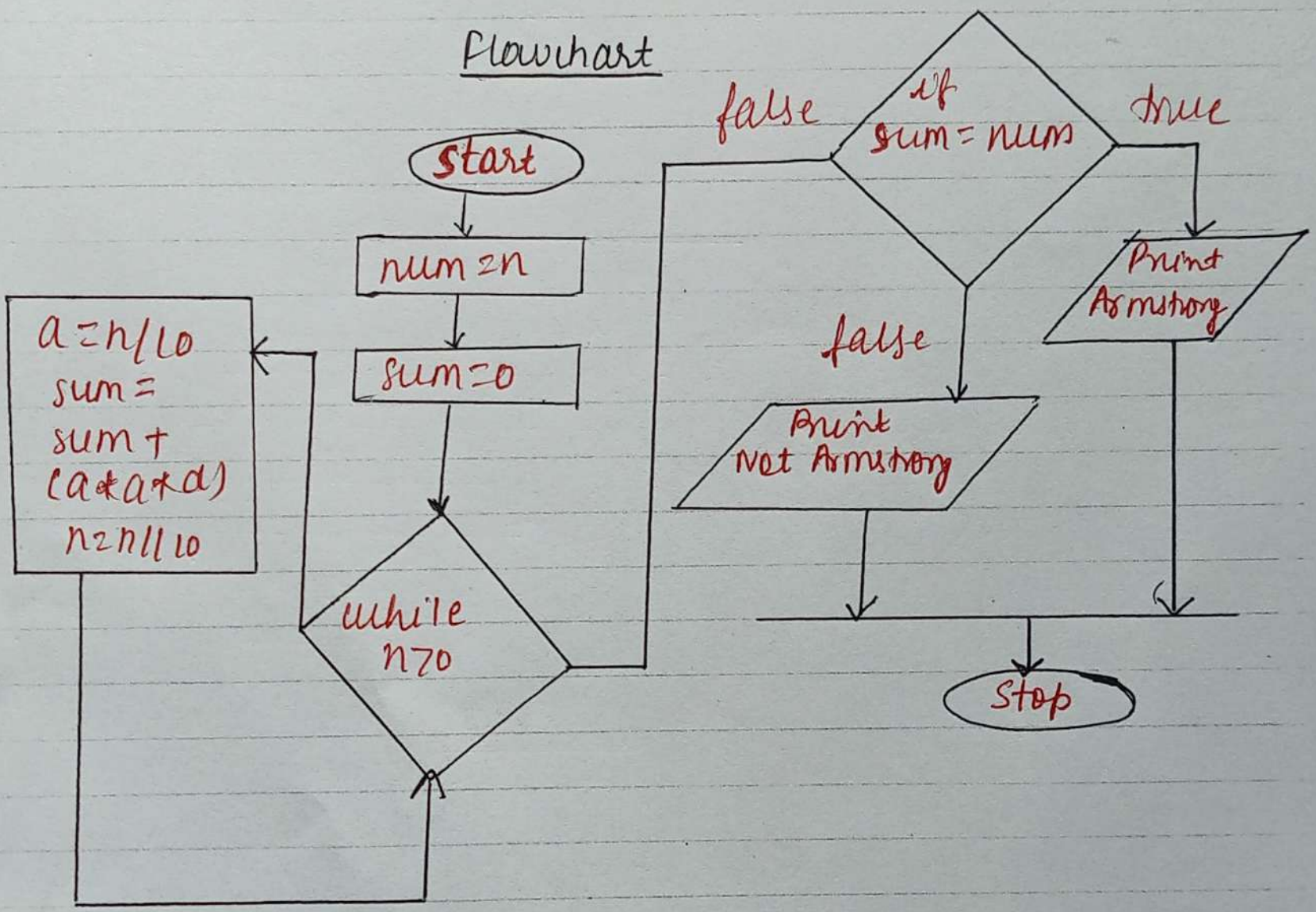
Algorithm

1. Start
2. enter a n digit number
3. num=n
4. rev=0
5. while n>0 , true then go to step 6 otherwise goto step 7
6. d=n%10
   sum= sum+ (d*d*d)
   n=n//10 , then goto step 5
7. if sum==num , true, then goto step 8 else goto step 9

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8. print armstrong & goto step 10
9. print not armstrong
10. stop

Flowchart



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Write a program to count the number of digit in a n digit number.

```
num = int (input ("enter the numbers"))
count = 0
while num > 0 :
        num = num // 10
        count = count + 1

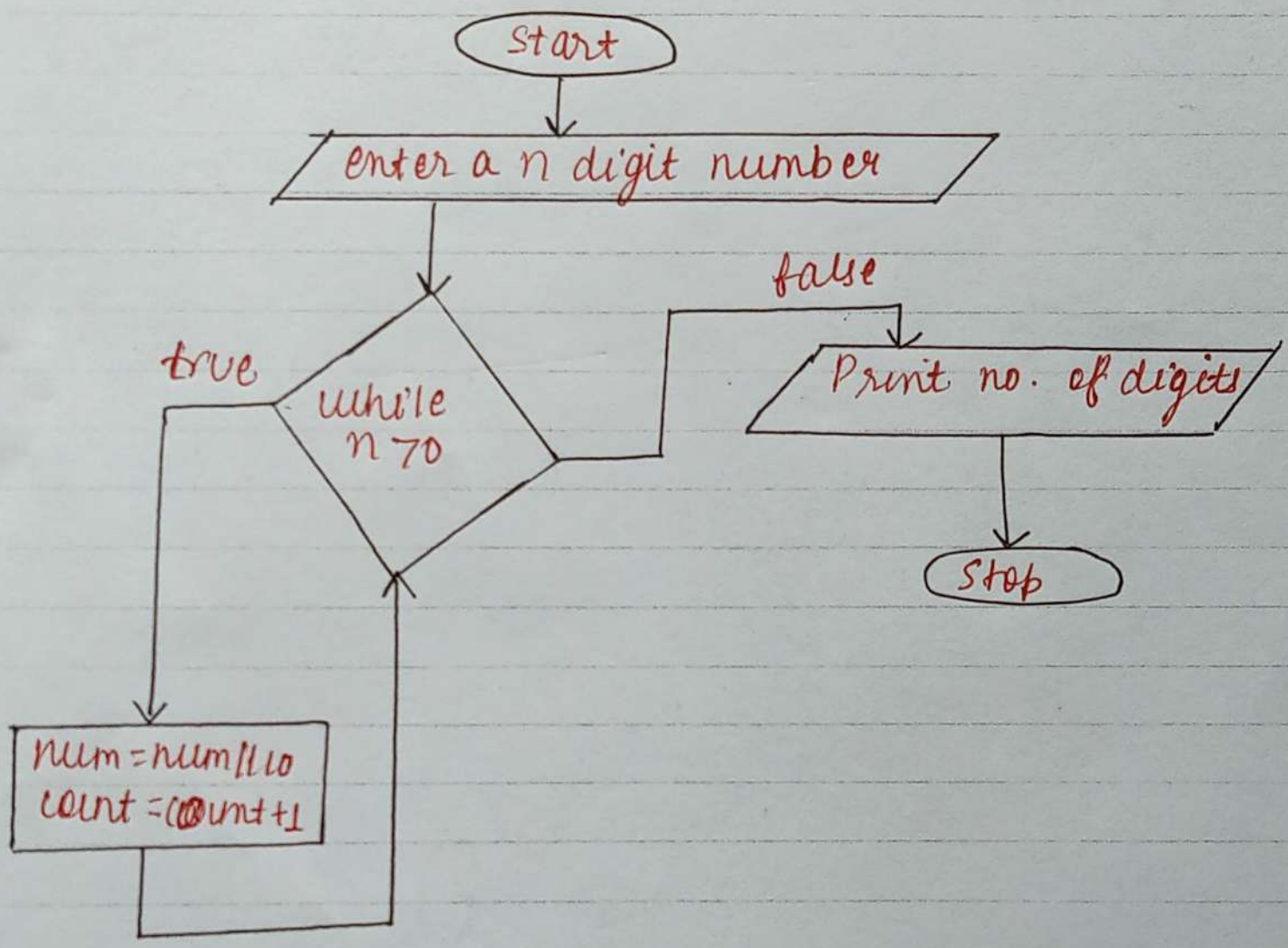
print ("Number of digits :", count)
```

Algorithm

1. Start
2. enter the n digit number
3. count = 0
4. while num > 0, then goto step 5, otherwise goto step 6
5. num = num // 10
   count = count + 1
6. Print no. of digit
7. stop

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flowchart

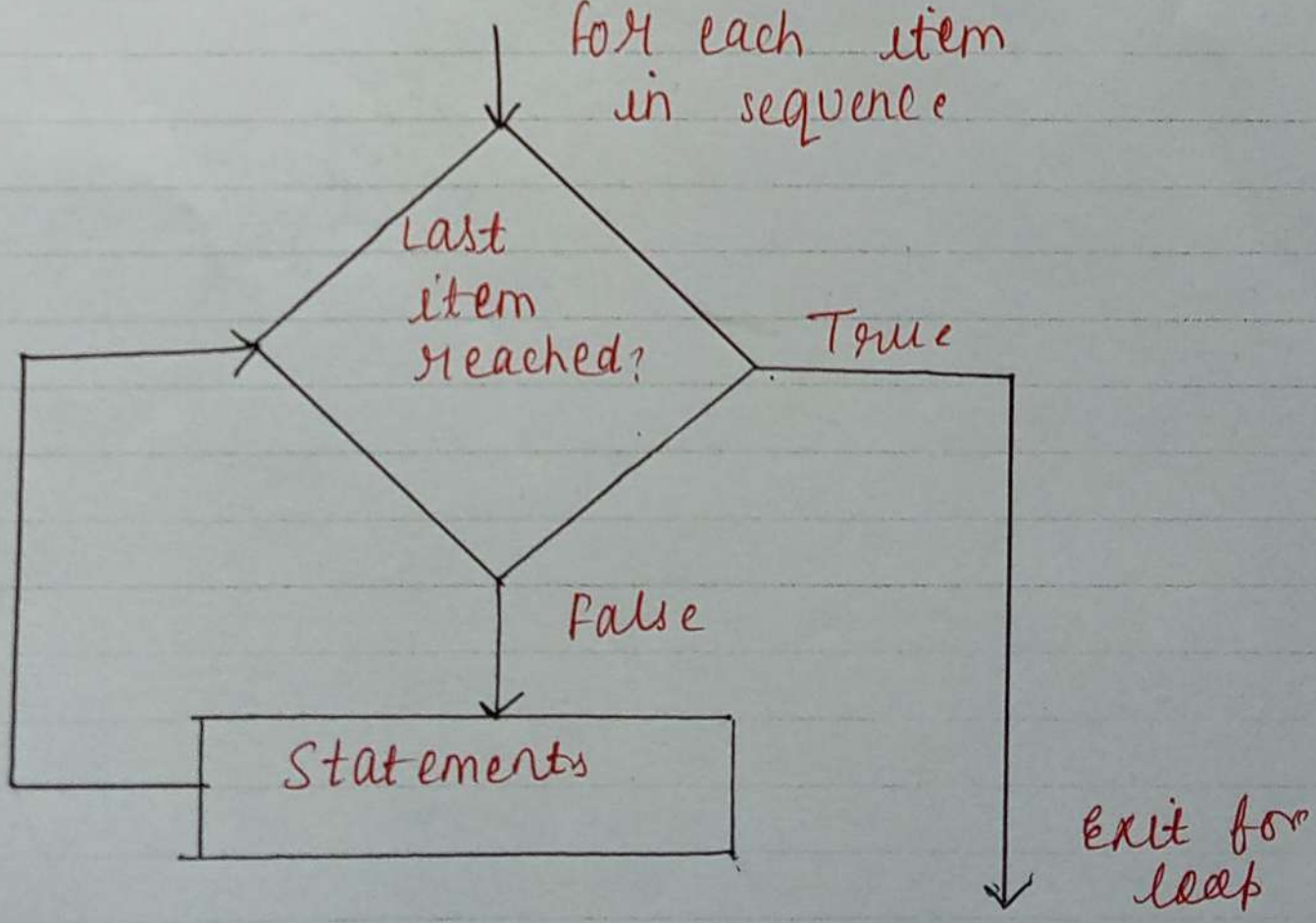


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For loop

- Python For loop is used for sequential traversal i.e. used for iterating over like string, Tuple, list, set or dictionary.

- For var in iterable:
- # statements
- Here the iterable is a collection of objects like lists, and tuples. The indented statement inside the For loops are executed once for each item in an iterable.
  The variable var takes the value of the next item of the iterable each time through the loop.



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For loop with list

```
l = ["gateway", "classes", "hello"]
for i in l:
    print(i)
```

Output:

gateway
classes
hello

For loop with dictionary

```
person = {"name": "John", "age": 30, "city":
          "New york"}
for key in person.keys():
    print(key)
```

Output:

name
age
city

```
person = {"name": "John", "age": 30, "city":
          "New york}
for value in person.values():
    print(value)
```

Output:

~~Name~~: John
~~age~~: 30
~~city~~: New york

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```
person = {"name": "John", "age": 30, "city":
                New York"}
    for key, value in person.items():
        print (key, ":", value)
```

output:

```
name: John
age: 30
city: New york
```

```
                For loop with string
print (" string Iteration")
    s = "python"
        For i in s:
            print (i)
```

output:

```
p
y
t
h
o
n
```

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Python for loop with a step size

This code uses a for loop in conjunction with the range() function to generate a sequence of numbers starting from 0, upto (but not including) 10, and with a step size of 2. For each number in the sequence, the loop prints its value using the print() function. The output will show the numbers 0, 2, 4, 6 and 8.

```
for i in range(0, 10, 2):
    print(i)
```

output:

```
0
2
4
6
8
```

Range()

The python range() function returns a sequence of numbers, in a given range. The most common use of it is to iterate sequences on a sequence of numbers using python loop.

```
for i in range(5):
    print (i, end=" ")
```

output

```
0
1
2
3
4
```

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Syntax of python ranges function

Syntax: range (start, stop, step)

Parameter:

start: [optional] start value of the sequence.

stop: next value after the end of the sequence

step: [optional] integer value, denoting the difference between any two numbers in the sequence.

Return: Returns an object that represents a sequence of numbers.

what is the use of the range function in python.

In simple terms, ranges allows the user to generate a series of number within a given range. Depending on how many arguments the user is passing to the function, the user can decide where the series of numbers will begin and end, as well as how big the difference will be between one number and the next.

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Python range() function takes can be initialised in 3 ways.

- range (stop) takes one argument
- range (start , stop) takes two arguments.
- range (start , stop , step) takes three arguments.

Python range(stop)

When the user call range() with one argument, the user will get a series of numbers that starts at 0 and includes every whole number up to, but not including, the number that the user has provided as the stop.

```
For i in range ( 6 ):
    print ( i, end = "" )
```

0 1 2 3 4 5

Python range (start, stop)

When the user call range() with two arguments, the user gets to decide not only where the series of numbers stops but also where it starts, so the user doesn't have to start at 0 all the time. Users can use range () to generate series of numbers from X to Y using range (X, Y).

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```
for i in range(5,20):
    print(i, end=" ")
```

output:

5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

Python range(start , stop ,step)

```
for i in range(0, 30, 4):
    print( i, end=" ")
```

output :

0 4 8 12 16 20 24 28

Incrementing the Range using a positive step

```
for i in range [0,30,4]:
    print (i, end=" ")
```

output:

0 4 8 12 16 20 24 28

Python range() using Negative step

```
for i in range( 25, 2,-2):
    print (i, end=" ")
```

output:

25 23 21 19 17 15 13 11 9 7 5 3

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For loop programs

Write a program to print the factorial of a number

```
num = int ( input (" enter the number\n"))
factorial=1
    if num < 0:
      print ( "sorry, factorial does not exist
              for negative numbers")
    elif num==0:
        print (" The factorial of 0 is 1")
    else:
      for i in range (1, num+1):
      factorial= factorial * i
      print (" The factorial of", num, "is", factorial)
```

Write a program to print the Fibonacci series upto n term

```
n terms= int ( input ( " How many terms ?"))
    n1, n2=0, 1
      count=0
    if nterms <=0:
      print (" Please enter a positive number")
    elif nterms==1:
      print (" Fibonacci sequence upto", nterms, ":")
        print (n1)
  elif nterms==2
      print (" Fibonacci series upto", nterms.":")
        print (n1)
         print (n2)
```

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```
else:
print (" Fibonacci series :")
    print ( n1)
     print (n2)
    for i in range (3, nterm +1):
      nth = n1 +n2
       print (nth)
         n1 = n2
         n2 = nth
```

Write a program to print the table of a number

```
number = int (input(" enter the number\n"))
print(" the multiplication table of ", number)
  for count in range (1, 11):
    print( number, " x ", count, "=", number * count)
```

Write a program to print the number of even numbers in a range

```
maximum= int ( input (" Please enter the
                      maximum value:"))
for number in range (1, maximum +1):
    if ( number %.2 == 0):

        print (" {0:}", format ( number))
```

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Write a program to count the number of even number in a range

```
maximum = int (input (" Please enter the
                      maximum value :"))
    count = 0
For number in range (1, maximum + 1):
    if (num %. 2 == 0):
        print (" {0y" . format (number))
        count = count + 1
    print (" total even number in range :",
                      count)
```

Write a program to display the n terms of even natural numbers and their sum.

```
maximum = int (input (" Please enter the maximum
                      value :"))
    total = 0
for number in range (1, maximum + 1):
    if (number %. 2 == 0):
        print ( "{0y". format( number))
        total = total + number

print ("The sum of even numbers from 1 to {0y
                = {1y". format (number, total))
```

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Write a program to print the number of odd number in a range

```
maximum = int (input (" Please enter the maximum value :"))
for number in range (1, maximum+1):
    if ( number %.2! =0):

        print ("%d", format (number))
```

Write a program to count the number of odd number in a range

```
maximum = int ( input (" Please enter the
                Maximum value :")).
count =0
for number in range(1, maximum+1):
    if (number %.2! =0):
        print ("%d". format (number))
        count = count +1

print (" total number of odd number
        in a range ", count)
```


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Write a program to display the n terms of odd natural numbers and their sum

```
maximum = int(input("Please enter the
                maximum value:"))
    total=0
for number in range(1, maximum+1):
  if (number %2 != 0):
      print("  {0}".format(number))
       total= total+ number

print("The sum of odd numbers from
    1 to {0} = {1}".format( number,
                            total))
```

Write a program to find out the Armstrong number within range.

```
low = int(input("enter low range:"))
  up  = int(input("enter up range:"))
   for num in range(low, up+1):
      sum=0
        temp= num
     while temp >0:
       digit = temp %10
       sum += digit **3
          temp//=10
       if num == sum:
          print(num)
```

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Write a program to find out the palindrome number within range

```
low = int ( input ("Enter low range:"))
up= int ( input (" Enter up range:"))
for num in range( low, up+1):
    rev =0
    temp= num
    while temp >0:
        digit = temp %10
        rev = rev * 10+ digit
        temp// =10
    if num == rev:
        print (num)
```

Break statement

The break statement is used to terminate the loop immediately when it is encountered

The syntax of the break statement is:

```
break;
```

```
for i in range(5):
    if i==3:
        break
    print(i)
```

output:

```
0
1
2
```

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```
for val in sequence:
    # code
    if condition:
        break
    # code
```

```
while condition:
    # code
    if condition:
        break
    # code
```

```
i = 1
while i <= 10:
    print('6 *', (i), '=', 6 * i)
    if i >= 5:
        break
    i = i + 1
```

output:

```
6 * 1 = 6
6 * 2 = 12
6 * 3 = 18
6 * 4 = 24
6 * 5 = 30
```

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continue statement

The continue statement is used to skip the current iteration of the loop and the control flow of the program goes to the next iteration.

The syntax of the continue statement is:

```
continue;
```

```
for i in range(5):
    if i==3:
        continue
        print(i)
```

Output:

```
0
1
2
4
```

```
for val in sequence:
    # code
    if condition:
        continue
    # code
```

```
while condition:
    # code
    if condition:
        continue
    # code
```

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```
num = 0
while num < 10:
    num += 1
    if (num % 2) == 0:
        continue
    print (num)
```

Output

1
3
5
7
9

Write a program to check whether a number is prime number or not

```
num = int (input( "enter the number \n"))
Flag = False
if num == 1:
print( num, " is not a prime number")
else:
for i in range (2, num):
    if (num % i == 0):
        Flag = True
        break
if flag == True:
print(" not prime")
else:
print (" prime")
```

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Pass statement

Python pass statement is a null statement. But the difference between pass and comment is that comment is ignored by the interpreter whereas pass is not ignored.

- SYNTAS
- PASS

what is pass statement in python?
when the user does not know what code to write, so user simply places a pass at that line. Sometimes, the pass is used when the user doesn't want any code to execute. So users can simply place a pass where empty code is not allowed, like in loops, function definition, class definitions, or in if statements. So using a pass statement user avoids this error.

Why python Needs "pass" statement?

If we do not use pass or simply enter a comment or a blank here, we will receive an Indentation error message.

```
a = 10
b = 20
if (a<b):
    pass
else:
    print("b<a")
```

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```
li = ['a', 'b', 'c', 'd',]
for i in li:
    if (i == 'a'):
        pass
    else:
        print (i)
```

Output :

b
c
d

Use of pass keyword in python class

The pass keyword can also be used in empty class in python.

```
class geekclass:
    pass
```

Purpose of loops

The purpose of loops is to repeat the same or similar code a number of times. This number of times could be specified to a certain number of number of times to be dictated by certain condition being met.

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Write a program to print the prime number within an interval.

```
lower = int ( input (" enter the upper
                    range \n"))

upper = int ( input (" enter the lower
                    range \n"))

print (" Prime numbers between ", lower,
        "and", upper," are :")

if num > 1 :
   for i in range ( 2, num):
   if (num % 1) == 0:
      break

else:
   print (num)
```


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Difference between for loop & while loop

for loop	while loop
For loop is used to iterate over a sequence of items.	while loop is used to repeatedly execute a block of statements while a condition is true.
For loops are designed for iterating over a sequence of items. Eg. list, tuple etc.	while loop is used when the number of iterations is not known in advance or when we want to repeat a block of code until a certain condition is met.
For loops require a sequence to iterate over.	while the loop requires an initial condition that is tested at the beginning of the loop.
For loop is typically used for iterating over a fixed sequence of items.	while loop is used for complex control flow situations.
For loop is more efficient than a while loop when iterating over sequences, since the no. of iterations is predefined.	While a loop may be more efficient in certain situations where the condition being tested can be evaluated quickly.

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Pattern Questions

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

```
n=5
# upper triangle
for i in range(n):
    for j in range(i+1):
        print('*', end=" ")
    print()
# lower triangle
for i in range(n):
    for j in (range(n-i-1):
        print('*', end=" ")
    print()
```

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```
    *
   * *
  * * *
 * * * *
* * * * *
```

```
size = 5
for i in range (size):
    for j in range (1, size - i):
        print (" ", end = " ")
    for k in range (0, i + 1):
        print ("*", end = " ")
    print ()
```

```
*
* *          → rows = 5
* * *
* * * *
* * * * *
```

```
rows = int ( input ("enter no of rows : "))
for i in range( rows):
    for j in range (i+1):
        print ("*", end = " ")
    print ("\n")
```


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```
1
1  2
1  2  3
1  2  3  4
1  2  3  4  5
```

→ rows = 5

```
rows = int ( input (" enter no. of rows:"))
  for i in range( rows):
      for j in range (i+1):
          print (j+1, end=" ")
      print( "\n")
```

```
A
B  B
C  C  C
```

→ rows = 3

```
rows = int (input(" enter the no. of rows:"))
  ascii_value = 65
    for i in range( rows):
        for j in range( i+1):
            alphabet= chr( ascii_value)
            print( alphabet, end=" ")

        ascii_value += 1
        print ("\n")
```

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```
* * * * *
* * * *
* * *
* *
*
```

→ rows = 5

```
rows = int ( input (" enter no. of rows: "))
    for i in range ( rows, 0, -1):
        for j in range(0,i ):
            print (" * ", end=" ")

    print("\n")
```

```
rows = int ( input (" enter no. of rows:"))
    for i in range ( rows, 0, -1):
        for j in range (1, i+1):
            print (j, end=" ")

    print("\n")
```

enter no. of rows : 4

```
1 2 3 4
1 2 3
1 2
1
```

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```
rows = int(input("enter no. of rows:"))
for i in range(rows,0,-1):
    for j in range(0,i):
        print("*", end=" ")

    print("\n")
```

enter no. of rows : 3

```
*   *   *
*   *
*
```

```
rows = int(input("enter no. of rows:"))
K=0
for i in range(1,row+1):
    for space in range(1,(rows-i)+1):
        print(end="")

    while k!=(2*i-1):
        print("*", end="")
        K+=1
    k=0
    print()
```

enter no. of rows : 4

```
      *
    * * *
  * * * * *
* * * * * * *
```

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For loop with tuple

A tuple is also a sequence of values just like a list. It is immutable and enclosed in parenthesis instead of square brackets.

```
items = ('one', 'two', 'three')
for item in items:
    print(item)
```

output:

```
one
two
three
```

For loop with list

```
Fruits = ['orange', 'apple', 'pear', 'banana',
          'kiwi']
for fruit in fruits:
    print(fruit)
```

output:

```
orange
apple
pear
banana
kiwi
```

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For loop with string

```
items = 'looping'
for item in items:
    print (item)
```

Output:

```
l
o
o
p
i
n
g
```

For loop with else

You can also use else keyword in python for loop. This is useful when you want to execute some code when the loop is finished.
The else block is executed when the loop is finished.

```
for i in range(3):
    print (i)
else:
    print("Loop finished")
```

Output:

```
0
1
2
Loop finished
```

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